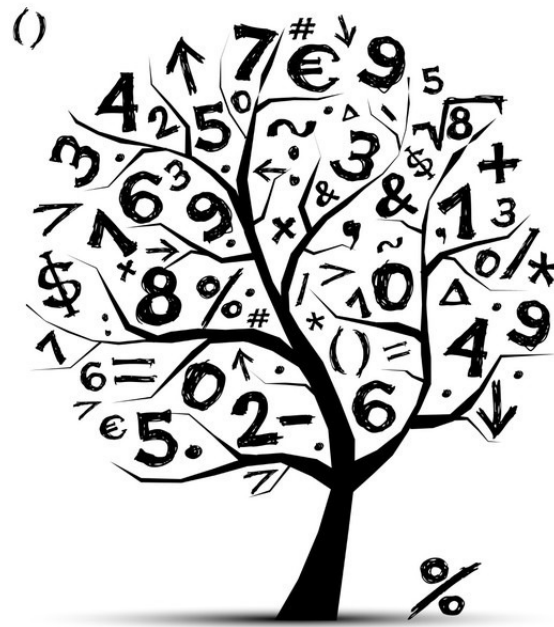


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Exercises: Matrix operations



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Exercise 1

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ -2 & 1 & 4 \end{pmatrix}.$$

Apply to A , one after the other, the elementary row operations $R_1 \leftrightarrow R_3$, $R_2 \leftarrow -2R_2$ and $R_3 \leftarrow R_3 + 2R_1$, and write the matrix obtained after each operation.

Each operation is applied to the matrix produced by the previous one:

$$\begin{pmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ -2 & 1 & 4 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 1 & 4 \\ 3 & 0 & 2 \\ 1 & 2 & -1 \end{pmatrix} \quad (R_1 \leftrightarrow R_3)$$

$$\begin{pmatrix} -2 & 1 & 4 \\ -6 & 0 & -4 \\ 1 & 2 & -1 \end{pmatrix} \quad (R_2 \leftarrow -2R_2)$$

$$\begin{pmatrix} -2 & 1 & 4 \\ -6 & 0 & -4 \\ -3 & 4 & 7 \end{pmatrix} \quad (R_3 \leftarrow R_3 + 2R_1)$$

Note that in the last step R_1 is the *current* first row $(-2, 1, 4)$: $(1, 2, -1) + 2(-2, 1, 4) = (-3, 4, 7)$.

Exercise 2

Consider the matrix

$$B = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}.$$

Apply to B , one after the other, the elementary column operations $C_1 \leftrightarrow C_3$, $C_2 \leftarrow C_2 + C_1$ and $C_3 \leftarrow 2C_3$, and write the matrix obtained after each operation.

$$\begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 1 \\ 1 & 3 & 0 \end{pmatrix} \quad (C_1 \leftrightarrow C_3)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 4 & 0 \end{pmatrix} \quad (C_2 \leftarrow C_2 + C_1)$$

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & 4 & 0 \end{pmatrix} \quad (C_3 \leftarrow 2C_3)$$

In the second step: $\begin{pmatrix} -1 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$.

Exercise 3

Let $A \in \mathbb{R}^{3 \times 3}$ with $\det(A) = 5$. Without computing any matrix entry, find the determinant of the matrix obtained from A by:

1. the row swap $R_1 \leftrightarrow R_2$;
2. the row scaling $R_2 \leftarrow -3R_2$;
3. the row addition $R_3 \leftarrow R_3 + 7R_1$;
4. the three operations above, applied one after the other;
5. the column swap $C_1 \leftrightarrow C_3$ followed by the column scaling $C_2 \leftarrow 4C_2$.

Finally, compute $\det(2A)$.

We use the effect of the elementary operations on the determinant: a swap changes the sign, a scaling by λ multiplies it by λ , an addition does not change it.

1. $\det = -\det(\mathbf{A}) = -5.$

2. $\det = -3\det(\mathbf{A}) = -15.$

3. $\det = \det(\mathbf{A}) = 5.$

4. $\det = (-1) \cdot (-3) \cdot 1 \cdot \det(\mathbf{A}) = 15.$

5. $\det = (-1) \cdot 4 \cdot \det(\mathbf{A}) = -20$ (column operations act as row operations).

Finally, $2\mathbf{A}$ is obtained by scaling each of the 3 rows by 2, so

$$\det(2\mathbf{A}) = 2^3 \det(\mathbf{A}) = 8 \cdot 5 = 40.$$

For instance, all these values can be checked on $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}$, which has $\det(\mathbf{A}) = 5.$

Exercise 4

Using elementary row operations (and no expansion formula), show that the following matrices have determinant equal to zero:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Row additions do not change the determinant, and a matrix with a zero row has determinant zero (Laplace expansion along that row).

For $\mathbf{A}$, the third row is twice the first one:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 0 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_1)$$

Hence $\det(\mathbf{A}) = 0$.

For $\mathbf{B}$:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 4R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 7R_1)$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 2R_2)$$

Hence $\det(\mathbf{B}) = 0$.

Exercise 5

Which of the following matrices are in row echelon form? Justify your answer.

$$\mathbf{M}_1 = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{M}_2 = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{M}_3 = \begin{pmatrix} 2 & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 5 \end{pmatrix}, \quad \mathbf{M}_4 = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

- **M_1 : yes.** The pivots are in columns 1 and 3 (strictly moving to the right) and the zero row is at the bottom.
- **M_2 : no.** The pivot of row 2 is in column 1, which is to the left of the pivot of row 1 (column 2).
- **M_3 : no.** The zero row (row 2) is above a nonzero row (row 3).
- **M_4 : yes.** The pivots are in columns 1 and 2 and the zero row is at the bottom.

Exercise 6

Reduce the following matrix to row echelon form with Gaussian elimination and indicate the pivots:

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \\ 1 & 2 & 2 & 4 \end{pmatrix}.$$

The first pivot is $a_{11} = 1$; we eliminate the entries below it, then we look for the next pivot:

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \\ 1 & 2 & 2 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 1 & 2 & 2 & 4 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

After the first step, the second column has no nonzero entry in rows 2 and 3, so we move to the third column: the second pivot is the entry 1 in position (2,3). The last matrix is in row echelon form, with pivots 1 (position (1,1)) and 1 (position (2,3)), and one zero row.

Exercise 7

Reduce the matrix

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}$$

to upper triangular form with Gaussian elimination, and use the result to compute $\det(A)$.

The multipliers are $\frac{4}{2} = 2$ and $\frac{8}{2} = 4$ for the first column, and $\frac{3}{1} = 3$ for the second column:

$$\begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 8 & 7 & 9 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 4R_1)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_2)$$

Only row additions were used ($s = 0$ row swaps), hence

$$\det(A) = 2 \cdot 1 \cdot 2 = 4.$$

Exercise 8

Compute the determinant of

$$A = \begin{pmatrix} 0 & 2 & 4 \\ 1 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$$

by Gaussian elimination. Check the result with the Laplace expansion along the first row.

Since $a_{11} = 0$, we start with a row swap:

$$\begin{pmatrix} 0 & 2 & 4 \\ 1 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 4 \\ 3 & 1 & 1 \end{pmatrix} \quad (R_1 \leftrightarrow R_2)$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & -2 & -5 \end{pmatrix} \quad (R_3 \leftarrow R_3 - 3R_1)$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & -1 \end{pmatrix} \quad (R_3 \leftarrow R_3 + R_2)$$

We used $s = 1$ row swap, hence

$$\det(\mathbf{A}) = (-1)^1 \cdot 1 \cdot 2 \cdot (-1) = 2.$$

$$\text{Check: } \det(\mathbf{A}) = 0 \cdot (1 - 2) - 2 \cdot (1 - 6) + 4 \cdot (1 - 3) = 10 - 8 = 2.$$

Exercise 9

Compute the determinant of the 4×4 matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 2 \\ 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 3 \end{pmatrix}$$

by Gaussian elimination.

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 2 \\ 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 3 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 1 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_1)$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 3 \end{pmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix} \quad (R_4 \leftarrow R_4 - R_2)$$

The matrix is upper triangular and only row additions were used, hence

$$\det(\mathbf{A}) = 1 \cdot 1 \cdot 1 \cdot 3 = 3.$$

(With the Laplace expansion we would need to compute four 3×3 determinants.)

Exercise 10

Reduce the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 1 & 3 \\ 4 & 4 & 2 \end{pmatrix}$$

to upper triangular form using Gaussian elimination with **partial pivoting**. Then compute $\det(\mathbf{A})$ and check that all the multipliers have absolute value at most 1.

Step 1: in the first column $|1| < |2| < |4|$: the pivot is 4, in row 3. **Step 2:** after the elimination, in the second column (rows 2 and 3) we have $|-1| < |2|$: the pivot is 2, in row 3.

$$\begin{pmatrix} 1 & 3 & 1 \\ 2 & 1 & 3 \\ 4 & 4 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 2 & 1 & 3 \\ 1 & 3 & 1 \end{pmatrix} \quad (R_1 \leftrightarrow R_3)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & -1 & 2 \\ 1 & 3 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - \frac{1}{2}R_1)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & -1 & 2 \\ 0 & 2 & \frac{1}{2} \end{pmatrix} \quad (R_3 \leftarrow R_3 - \frac{1}{4}R_1)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & 2 & \frac{1}{2} \\ 0 & -1 & 2 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

$$\begin{pmatrix} 4 & 4 & 2 \\ 0 & 2 & \frac{1}{2} \\ 0 & 0 & \frac{9}{4} \end{pmatrix} \quad (R_3 \leftarrow R_3 + \frac{1}{2}R_2)$$

The multipliers are $\frac{2}{4} = \frac{1}{2}$, $\frac{1}{4}$ and $\frac{-1}{2} = -\frac{1}{2}$: all have absolute value at most 1, because each pivot is the entry of largest absolute value in its column (on or below the diagonal).

We used $s = 2$ row swaps, hence

$$\det(A) = (-1)^2 \cdot 4 \cdot 2 \cdot \frac{9}{4} = 18.$$